Chris Cameron

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AI-native researcher, developer, and mechatronics engineer. PhD researcher in electrophysiology (completion expected early 2027) and creator of Cubie. Part-time lecturer in mechatronics engineering, including computer architecture, C programming, circuit design, and embedded systems.

Cubie — GPU numerical integration

github.com/cubiepy/cubie

I built Cubie, an open-source Python ODE solver for consumer NVIDIA GPUs, to support electrophysiology simulations for a real-time application. Current benchmarks (draft, unpublished) show **10,000× speedup** over CPU implementations (SciPy, MATLAB) and **100×** over the nearest GPU competitor, DiffEqGPU.jl, on real-world problems.

Once CUDA-shaped algorithms were in place and tiling and memory placement were addressed, further speedups were achieved by:

- using NVIDIA profiling tools to trace bottlenecks and stalls to individual instructions;
- rolling/unrolling loops based on measured instruction counts against the instruction cache;
- decreasing occupancy to reduce L2 cache misses (this one was surprising!);
- ordering and fusing operations at a whole-kernel level to reduce live ranges and reduce spilling; and
- placing buffers heuristically to trade off guaranteed-L1 shared memory against possible-L2/VRAM local memory based on working set size.

Cubie goes fast, but it is a scientific tool. The codebase is complex and correctness is critical. To maintain correctness, I had to:

- test numerical results against known references and custom CPU harnesses;
- review every pull request line-by-line (AI agents are prolific but often incorrect); and
- build approval processes including a ratcheting benchmark per PR and automated correctness checks.

Open-source contributions

numba-cuda / numba-cuda-mlir

Many merged and open pull requests address compiler performance, typing and lowering correctness, AST transforms, and kernel performance controls.

SymPy and lcapy — minor edits.